

# STANCEPARTS

## AIR CUP LIFT SYSTEM



## USER MANUAL

### IMPORTANT:

Read this manual carefully before installing or using the product.  
It is expected that users follow all guidelines to ensure safe and correct usage.  
**Save this manual for future reference.**



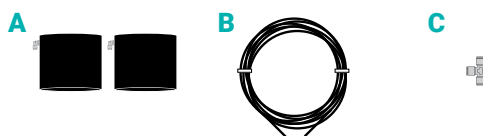
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### 1. What's in the box

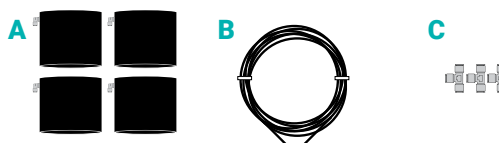
**1 pair of air cups**  
Part No. 1002

**A:** Air cup (2pcs)  
**B:** Airline 6mm  
**C:** T-splitter



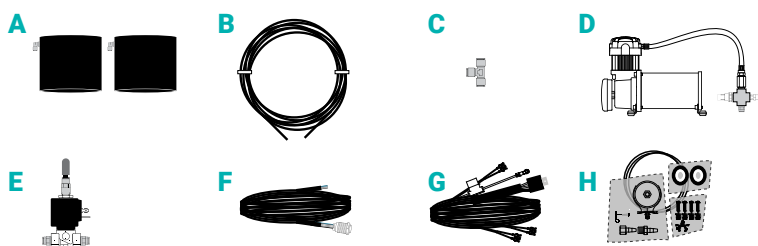
**2 pair of air cups**  
Part No. 1004

**A:** Air cup (4pcs)  
**B:** Airline 6mm  
**C:** T-splitter (3pcs)



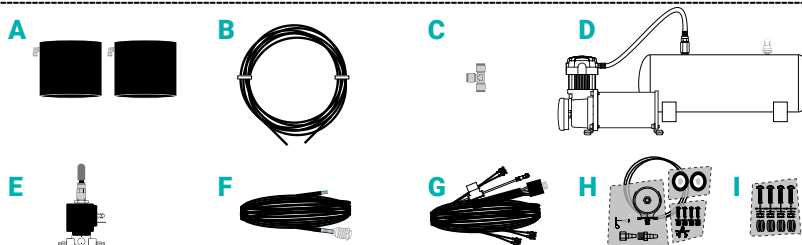
**Tankless front kit**  
Part No. 2002

**A:** Air cup (2pcs)  
**B:** Airline 6mm  
**C:** T-splitter  
**D:** 12v Compressor  
**E:** 3-way valve  
**F:** 16mm LED switch  
**G:** Wiring harness  
**H:** Compressor assembly kit (4pcs)



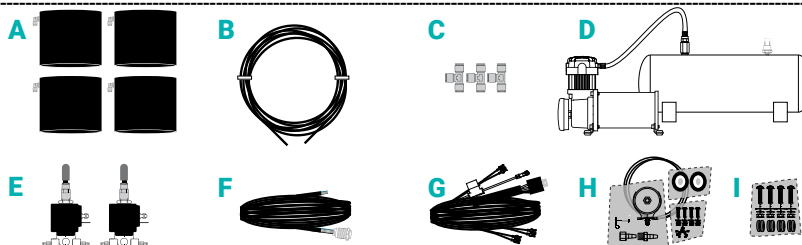
**Complete front kit**  
Part No. 3002

**A:** Air cup (2pcs)  
**B:** Airline 6mm  
**C:** T-splitter  
**D:** 12v Compressor setup  
**E:** 3-way valve  
**F:** 16mm LED switch  
**G:** Wiring harness  
**H:** Compressor assembly kit (4pcs)  
**I:** Tank installation kit



**Complete front + rear kit**  
Part No. 3004

**A:** Air cup (4pcs)  
**B:** Airline 6mm  
**C:** T-splitter (3pcs)  
**D:** 12v Compressor setup  
**E:** 3-way valve (2pcs)  
**F:** 16mm LED switch  
**G:** Wiring harness  
**H:** Compressor assembly kit (4pcs)  
**I:** Tank installation kit



## 2. Fitment check

**Double-check if the air cups fit your coilovers before installation.**

### Springs

The air cups can only be installed on linear springs, they will not fit on shaped springs, as shown in the figure.

- Maximum spring outer diameter: 98mm (89mm for the smaller 110mm cups)
- Minimum spring inner diameter: 60mm. (57mm on request)



### Ride Height

The air cup adds between 20 and 30mm in height. The air cup itself is 20mm thick, and with most coilovers, the top hat sits on top of the air cup, adding approximately 10mm.

If there is enough thread left on the coilovers to lower the springs by 30mm, there won't be any issues. If not, shorter springs will be required.



### Clearance

Ensure there is enough clearance around the coilovers, the air cups are 110 or 120mm in diameter, the fitting is 17mm.



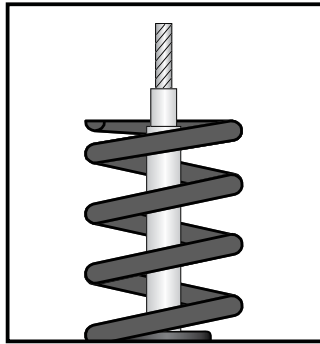
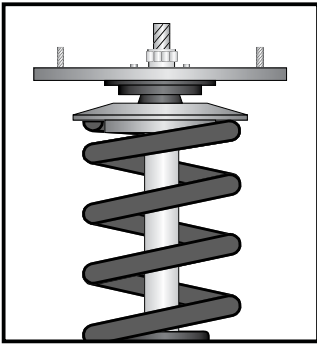
### Damper shaft

Before installing, verify that you have the correct size air cups. Measure the thickest part of the shaft using a caliper, below any rings or clips.

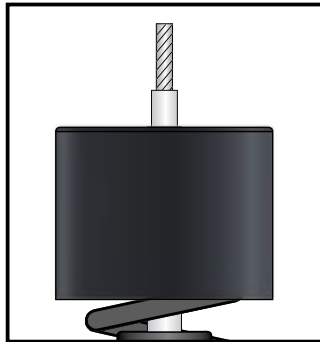
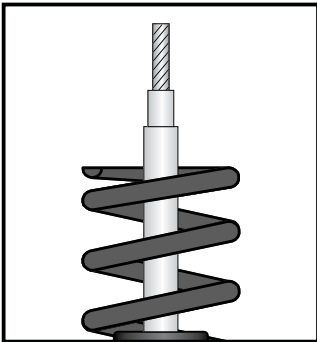
**Ordered the wrong size air cups?** Handle them with care, if they are in new condition without any marks we can swap them for the right size without any charge. You only need to pay for shipping.



### 3. Coilover assembly

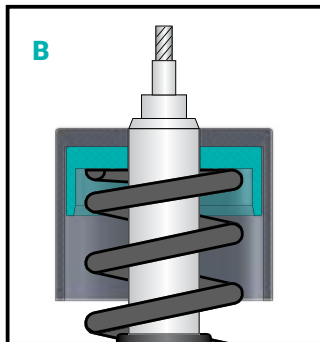
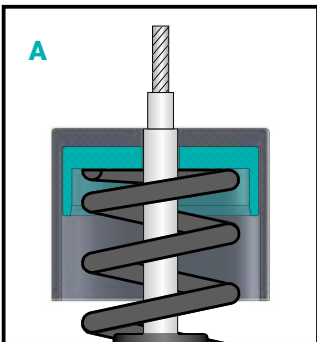


**Step 1:** Disassemble all the parts above the spring and keep them in order. If present, remove any washers and clips from the shaft. The air cup will be installed below these parts.



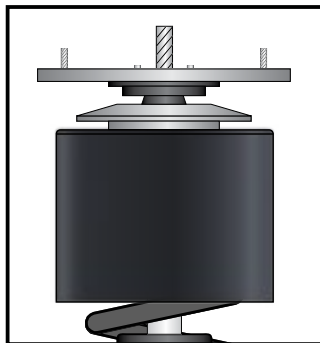
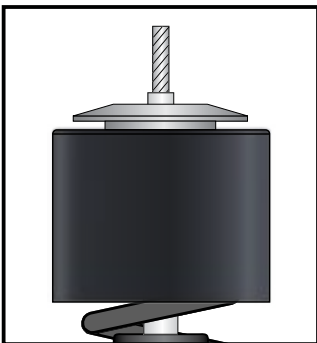
**Step 2:** Lower the spring by 20-30mm to install the air cup.

**Optional:** Cut the bump stops to allow more damper travel. We recommend keeping at least 15mm of bump stop.



**Step 3:** Carefully place the air cup over the shaft and ensure it sits on the thickest part of the shaft. Turn up the spring until the top surface of the air cup sits flush with the thickest part of the shaft.

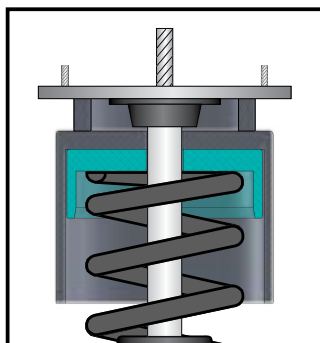
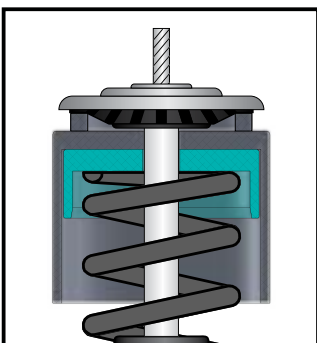
For inverted dampers (image B), make sure the air cup sits all the way down on the thickest part of the shaft.



**Step 4:** Ensure the tophat/top mount covers all the upper parts of the shaft and sits directly on the air cup. If there is a gap, it must be filled with spacers.

Now, reassemble all parts in reverse order on top of the air cup.

**For MacPherson struts:** double-check that the top mount can rotate freely.



**Different style topmounts:** The most important thing when using different style top mounts or OEM top mounts is to ensure the air cup rests on the same surface where the spring originally sat. In most cases, an adapter or spacer is needed to fill the gap between the air cup and the top mount.

**For MacPherson struts:** double-check that the top mount can rotate freely.

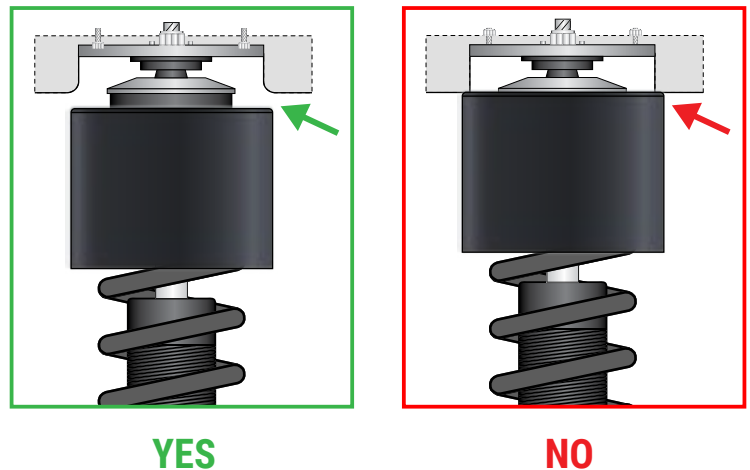


## 4. Air cup clearance

If you have MacPherson (rotating) struts in the front, keep in mind that the air cups will rotate along with the lower coilover body when steering. This does not apply to double wishbone suspension.

Ensure that the air cups can rotate freely without making contact with the chassis in any way. You can increase clearance by adding a spacer on top of the air cup.

**Regularly check that the air cups do not twist. If they do, the clearance is too small.**

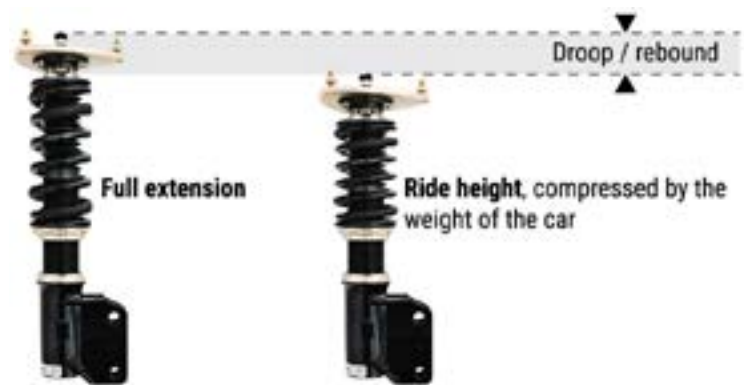


## 5. Spring adjustment

The lift from the air cups requires droop from the coilovers. If there is not enough droop, the air cups will not be able to lift the full 50mm.

As a side effect, when the air cups cannot lift the full 50mm, the remaining stroke will compress the spring. Depending on the amount, this can result in a stiffer ride when aired up.

There is a quick way to check how much droop you have: Jack up the car until the wheels come off the ground. Measure the space gained between the wheel and fender. This is the amount of droop you have, and therefore the amount of lift you will get (up to a max. of 50mm).



### What is droop?

Droop, also called rebound, is the amount of travel between ride height and full extension. Too much preload will reduce droop.

**If you are using extremely high spring rates, the car's weight may not compress the springs enough to create sufficient droop. This can be solved by installing helper springs, which provide the necessary droop for the air cups to lift.**

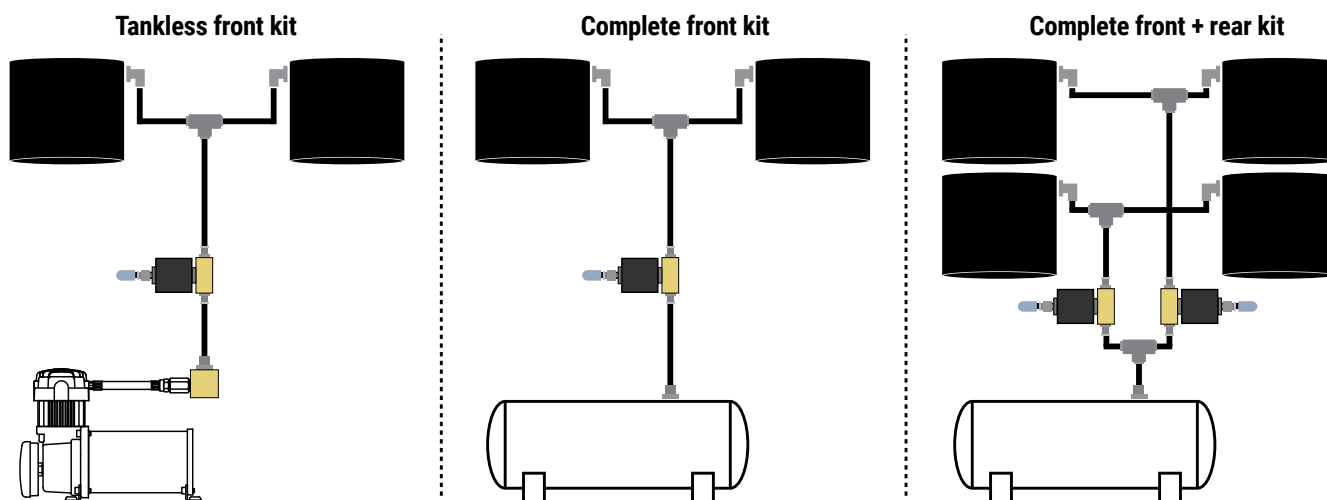
## Custom rear cups



**Separated spring and damper:** For a limited selection of cars with a separated spring and damper setup on the rear, we offer custom rear cups.

These will take up 30mm of height. Adjust the spring height adjuster by 30mm or install shorter springs, and place the air cup between the spring and the height adjuster.

## 6. Installation diagram



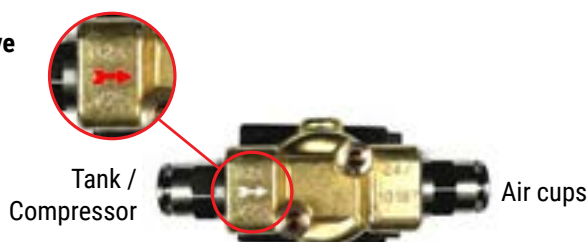
### Mounting the compressor and tank

Both the compressor and tank must be mounted upright. Choose a location that is as cool as possible and away from heat sources. The compressor is water and dust resistant but not waterproof or dustproof. Do not mount it in a place where it may come into contact with water or dirt.

**Cut the airlines straight and make sure they are firmly connected to the fittings. This prevents them from coming loose or leaking.**

### Connecting the 3-way valve

Bottom view



## 7. Wiring

**Follow the labels on the supplied wiring harness to connect all components correctly.**

- Battery:** Connect to the positive 12V battery terminal.
- Compressor:** Connect to the positive compressor wire. Connect the black compressor wire to ground.
- Ground:** Connect to a suitable chassis ground.
- Valve:** Connect to the 2 vertical pins on the valve. For front-only kits, use only one loom.
- Pressure switch:** Connect to the pressure switch on the tank, or to the manifold on tankless kits.
- Keyed power:** Connect to a power source that is only live when the ignition is on.
- Switch:** Connect the 3 switch wires color to color.

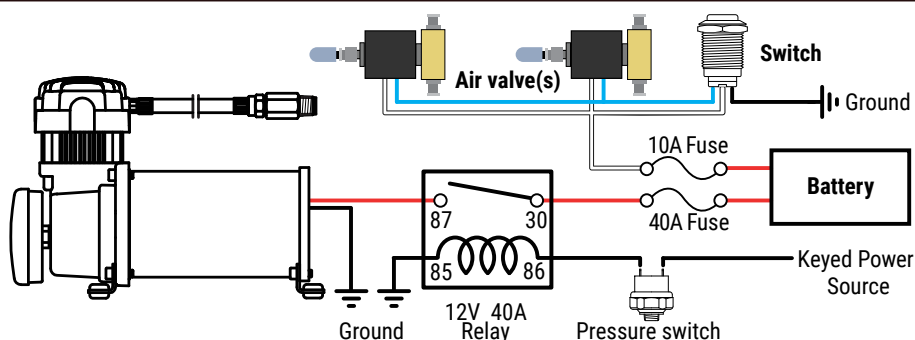


The valve is not polarity-sensitive, wire colors do not matter.

**Optional:** Second switch for separate front and rear control. Connect the black and white wires to the quick connector, same as the first switch. Connect the blue wire from the second switch to one valve, replacing the existing blue connector. That valve will then be controlled separately by the second switch.

### Electric diagram

If you choose to wire the system manually, please follow this diagram.



## 8. Operating instructions

**The system must be operated as described below.**

- The system should only be used when passing obstacles, not constantly while driving.
- Use only at speeds below 60 km/h / 40 mph and pass obstacles at an appropriate speed.
- Only operate when mounted on the car. Keep hands and fingers away from moving parts.
- The system should only be used in two modes: not in use (no pressure) and fully lifted (6-10 bar / 90-150 psi).
- **Max operating pressure for air cups, airlines, fittings, compressor, tank, and valves: 150 psi.**

## 9. Maintenance

### Compressor and tank

- Check the air filter every 3 to 6 months and replace if needed.
- Drain the tank every month by opening the tap until all moisture is removed.
- The tank must be replaced every 2 to 5 years from the date of purchase.
- Important: Regularly inspect the tank for rust. If any rust is found, replace it immediately.

**Bleed pressure from tank before servicing or adding attachments.**

## 10. Troubleshoot guide

**Air cups don't lift:** Turn off the system, disconnect the airlines from the air cups, and turn the system on. The airlines should release pressure. If no pressure is released, the system is either not pressurized or the valves are not opening. If the air cups have pressure but do not lift, the coilover assembly is incorrect, preventing the air cups from expanding.

**Compressor doesn't turn on:** 1. It might be overheated, let it cool down for 30 minutes. 2. Check if the fuse is blown. 3. Disconnect the two connectors from the pressure switch and hold them together. If it turns on, the pressure switch needs to be replaced. 4. If it still doesn't work, apply 12V directly to the compressor wires. If it doesn't turn on, the compressor needs to be replaced.

**Compressor starts running without using the air cups:** There is likely a leak somewhere between the compressor setup and the valves. Check all fittings and valves by spraying soapy water on them.

**Compressor runs continuously causing safety valve to open:** This could be caused by a faulty pressure switch, a failing safety valve, or incorrect wiring. Disconnect the two connectors from the pressure switch, if the compressor turns off, the wiring is correct. Start by replacing the pressure switch, then the safety valve if needed.

**Switch doesn't work:** 1. Check all fuses in the wiring harness: one 40A fuse near the battery and one 10A fuse near the relay. 2. Test by bypassing the switch: Disconnect the switch from the wiring harness. Use a short wire to connect the black and blue connectors in the wiring harness. If the system lifts, the switch is faulty. If not, the issue is elsewhere in the system.

## 11. Warranty

2-year warranty on the air cup, valves and other parts.

1-year warranty on the compressor setup.

### Warranty terms:

- Proof of purchase required.
- Product must be used properly according to the instructions.
- Proper installation must be done according to this manual.

### LIMITATION OF LIABILITY

Stanceparts B.V. is not liable for any damage, injury, or loss resulting from the installation, use, or misuse of this product. The buyer assumes all risks associated with its use, including but not limited to, vehicle damage, personal injury, or legal non-compliance. Stanceparts B.V. makes no guarantees regarding the suitability of this product for any particular purpose. Proper installation by a qualified professional is required. Failure to follow installation and usage instructions may result in malfunction, injury, or property damage. Stanceparts B.V. is not responsible for any incidental or consequential damages. This product may not be legal for use on public roads in some regions. Always check and comply with local laws and regulations before installation and use.



# #STANCEPARTS

**We'd love to see more about your project!**

Feel free to send us some pictures or videos to share on our social media channels.

Reach out to [@stanceparts on Instagram](#) or email [info@stanceparts.com](mailto:info@stanceparts.com)



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